

Question 41 (4 marks)

In a particular city, the heights of adult females and the heights of adult males are each normally distributed.

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Information relating to two females from that city is given in Table 1.

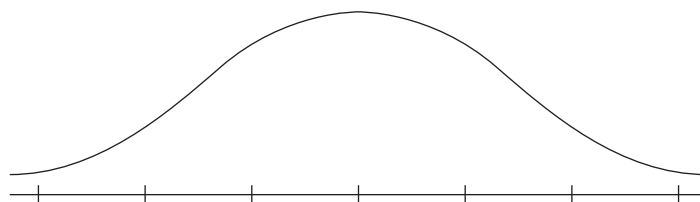
Table 1	<i>Height</i>	<i>Gender</i>	<i>Percentage of females in this city shorter than this person</i>
	175 cm	Female	97.5%
	160.6 cm	Female	16%

The means and standard deviations of adult females and males, in centimetres, are given in Table 2.

Table 2		<i>Mean</i>	<i>Standard deviation</i>
	Females	μ	σ
	Males	1.05μ	1.1σ

A selected male is taller than 84% of the population of adult males in this city.

By first labelling the normal distribution curve below with the heights of the two females given in Table 1, calculate the height of the selected male, in centimetres, correct to two decimal places.



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